

Analysis and Prediction of Consumer Financial Complaints

1. Project Statement

The Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database provides information about problems consumers experience with financial products and services. In addition to structured fields such as product, issue, company, state, and company response, many records include a free-text narrative in which consumers describe what happened in their own words. The CFPB notes that these narratives are consumers' descriptions of their experiences and that the database is not a statistical sample of all consumers in the financial marketplace (Consumer Financial Protection Bureau, n.d.).

Prior research also suggests that these narratives provide useful information beyond structured complaint categories. Foohey (2017), for example, analyzed 5,000 CFPB complaint narratives and found that consumers use the narrative field to communicate not only the financial problem itself but also frustration, fear, and the effects of these experiences on their lives. Other research has used NLP methods to identify systematic patterns and anomalies within CFPB complaint narratives (Gao et al., 2023).

The goal of our project was to use multiple data science methods to better understand consumer financial complaints. We focused on four related questions: What problems are consumers describing? Can complaint narratives reveal patterns beyond the CFPB's existing categories? How does complaint activity vary across geography and time? Finally, can historical complaint information be used to forecast complaint volume or predict company response outcomes?

The primary audience for this project is analysts, researchers, and regulators who use consumer complaint data to identify patterns, monitor trends, and better understand complaint outcomes.

To address these questions, we combined natural language processing (NLP), unsupervised learning, geographic analysis, time-series forecasting, and supervised machine learning. Each method answered a different part of the overall research question.

2. Data and Overall Methodology

The Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database is a publicly available collection of complaints related to financial products and services, including credit reporting, mortgages, credit cards, bank accounts, student loans, and debt collection. The database includes structured fields such as product, issue, company response, state, and ZIP code, as well as consumer-written complaint narratives when consumers consent to publication.

All analyses used data from the CFPB Consumer Complaint Database, although different subsets were needed for different research questions.

The 2025 narrative dataset contained 1,221,970 complaints. After preprocessing and filtering for the NLP analysis, 1,056,568 narratives remained. Credit reporting or other personal consumer reports represented 912,758 complaints, or about 75% of the 2025 narrative data, making credit reporting an important area for deeper analysis.

For the unsupervised analysis, we began with a stratified sample of 10,000 complaints from a prepared dataset containing more than one million narratives. After removing duplicates, very short submissions, heavily masked records, and highly repetitive openings, 4,727 complaints remained. Of these, 2,902 credit-reporting complaints were used for clustering.

The geographic and forecasting analysis required a longer historical period, so it used a 25% sample of CFPB complaints from 2011 through 2026. This provided coverage across the full study period while reducing the computational demands of working with the complete historical dataset. Because this was a sample, raw complaint totals in this part of the project should not be interpreted as the complete number of CFPB complaints.

The supervised-learning analysis returned to the 2025 narrative dataset and used a 70% training partition and 30% holdout partition. Only the training observations were used to fit the models, while the holdout data were reserved for evaluation.

3. Understanding Complaint Narratives

Sentiment and Financial Severity

We first used VADER sentiment analysis to examine the tone of complaint narratives. VADER is a rule-based sentiment model developed by Hutto and Gilbert (2014), originally with a focus on social-media-style text.

However, our results showed that sentiment should not be treated as a direct measure of financial severity. Complaints describing fraud, identity theft, or financial hardship can still contain neutral or positive wording. We therefore treated VADER as a measure of narrative tone rather than consumer harm or resolution quality.

To identify more domain-specific signals, we developed a financial severity framework using spaCy's PhraseMatcher and keyword dictionaries. Critical indicators included terms related to foreclosure, bankruptcy, lawsuits, garnishment, eviction, and hardship. High-severity indicators included fraud, identity theft, phishing, unauthorized transactions, and account takeover, while medium-severity indicators included issues such as incorrect reporting, billing errors, and overdraft fees.

The composite severity score assigned five points for each critical match, three for each high-severity match, and one for each medium-severity match. Sensitivity tests with alternative weighting schemes produced similar overall patterns, suggesting that the main results were not strongly dependent on the exact weights.

This gave us a better way to identify potentially serious financial issues than sentiment alone. However, the score should be viewed as a domain-informed indicator, not as a validated measure of actual consumer harm.

For this part of the analysis, we considered the severity framework useful if it could identify financial-risk patterns that general sentiment did not capture and if the overall results stayed similar when we changed the scoring weights. Since there is no ground-truth severity label in the CFPB data, we evaluated the framework using sensitivity testing rather than prediction accuracy.

The NLP analysis also created a binary relief indicator in which monetary or non-monetary relief was coded as 1 and other response categories were coded as 0. This variable was used for the NLP analysis and is different from the supervised-learning target described later.

Unsupervised Topic Analysis

Severity scoring identifies potentially serious language, but it does not tell us what specific types of problems consumers are describing. Because credit reporting accounted for most 2025 narratives, we applied unsupervised learning to credit-reporting complaints.

Each narrative was converted into a 768-dimensional embedding using the all-mpnet-base-v2 sentence-transformer model. Sentence embeddings were chosen because they represent semantic similarity rather than relying only on shared words. This general approach is supported by Sentence-BERT, which was developed to generate semantically meaningful sentence representations suitable for similarity comparison and clustering (Reimers & Gurevych, 2019).

UMAP reduced the embeddings from 768 dimensions to five dimensions. The selected configuration used 10 neighbors and cosine distance, followed by HDBSCAN with a minimum cluster size of 50 and min_samples of 20. Unlike methods that require the number of clusters to be selected beforehand, HDBSCAN could also leave narratives unassigned when they did not fit a sufficiently dense group.

We evaluated clustering based on interpretability, topic balance, outlier rate, representative complaints, topic keywords, and stability across random seeds. The selected $n_neighbors = 10$ configuration produced 12–13 topics across five seeds, with pairwise Adjusted Rand Index values from 0.716 to 0.773 and a mean of 0.740.

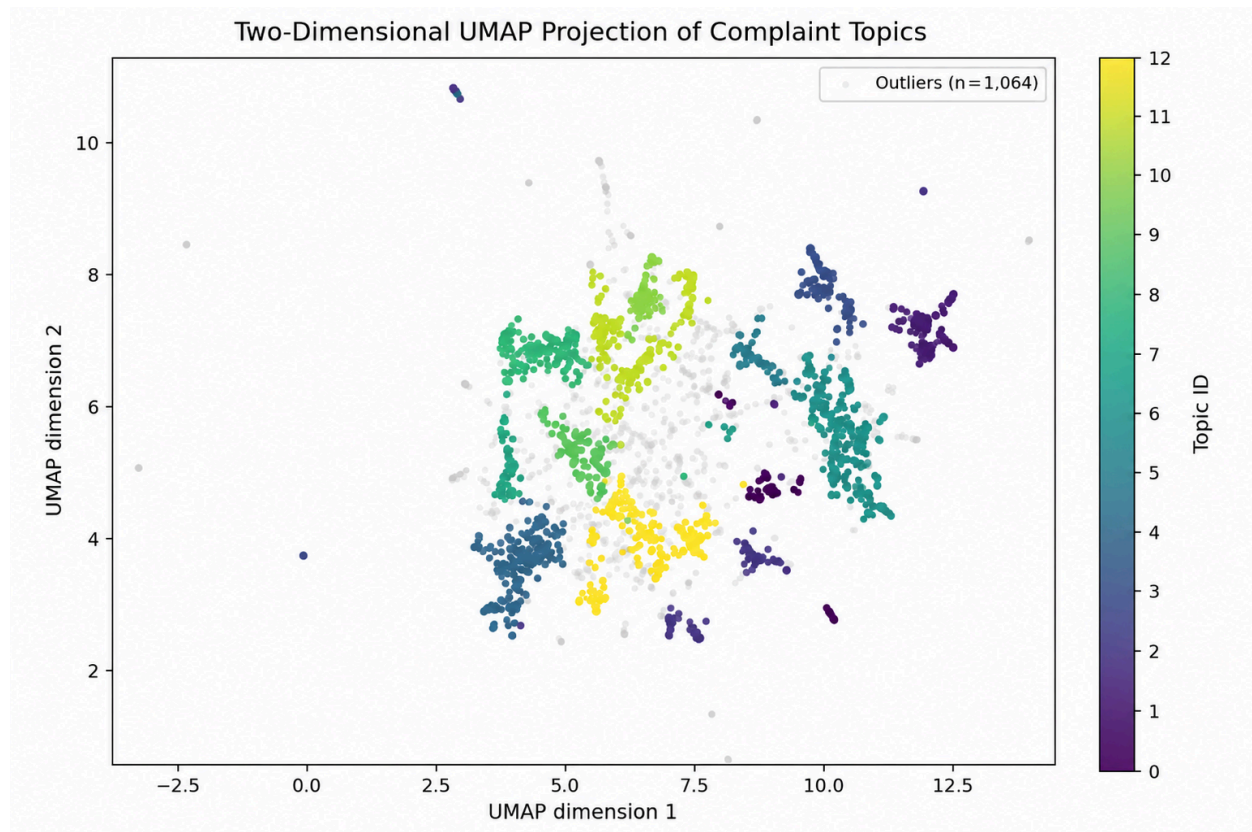
Sensitivity testing also showed why the final configuration was selected. Using Euclidean instead of cosine distance produced only five topics and placed 72.1% of complaints into one dominant cluster. Removing common legal and FCRA language caused the model to collapse into two broad topics, with 91.3% of complaints in one cluster.

The final model identified 13 topics. Of 2,902 complaints, 1,838 were assigned to a topic and 1,064 were classified as outliers, giving a 36.7% outlier rate. The largest topics included Identity Theft and Fraudulent Accounts, Late-Payment Reporting Errors, Account and Personal-Information Inaccuracies, and Multi-Bureau Reinvestigation Failures.

The clusters added detail to the CFPB's structured categories. For example, identity-theft complaints separated into general fraudulent accounts, formal FCRA Section 605B blocking requests, and cases where identity-theft accounts remained after documentation was submitted.

Investigation complaints also separated into Equifax-specific failures and broader multi-bureau reinvestigation problems.

Figure 1. Two-dimensional UMAP projection of credit-reporting complaints.



The figure displays the 13 topics and complaints classified as outliers. Some groups are relatively distinct while others overlap, which is reasonable because a single complaint can involve several related issues. The visualization is a two-dimensional summary; the actual clustering was performed on the five-dimensional UMAP representation.

4. Geographic and Temporal Patterns

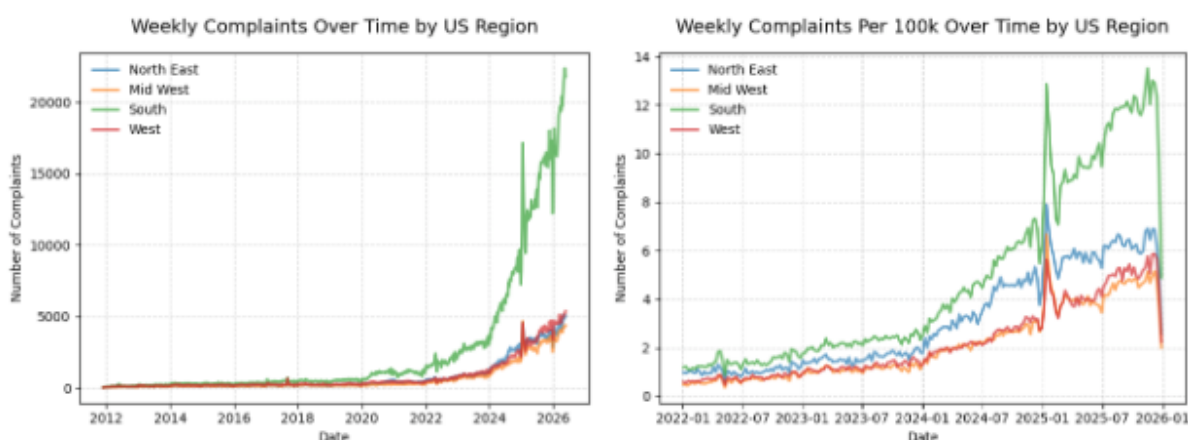
The narrative analyses focused mainly on what consumers reported. We next examined where and when complaints occurred.

ZIP codes were cleaned and geocoded using pgeocode, resulting in coordinates for approximately 3.6 million of 4 million ZIP-code observations.

Comparing raw complaint counts across regions can be misleading because more populated areas have more potential complainants. The CFPB specifically recommends considering population when comparing complaint volume across states or ZIP codes (Consumer Financial Protection Bureau, n.d.). We therefore grouped states into regions outlined by the US Census and normalized complaint counts as complaints per 100,000 residents.

After normalization, the South had a higher complaint rate than the other regions. We wondered whether regional differences in household credit conditions could be related to this pattern, since prior work has found lower average credit scores in Southern states (Woodard, 2025). However, we did not test this explanation, so more analysis would be needed to know whether it actually contributes to the higher complaint rate in the South.

Figure 2. Weekly complaint volume and population-adjusted complaint rates by U.S. region.



The time series also showed long-term growth and repeating seasonal patterns. Complaint volume generally increased near the beginning of the year and declined toward the end of the year. We report these as observed patterns rather than assuming a specific cause.

Complaint activity also changed sharply around 2025, during a period of substantial institutional change at the CFPB. A Congressional Research Service report documents that legislation enacted in 2025 reduced the CFPB's annual Federal Reserve funding cap by approximately 46% (Schneider, Tufts, & Webel, 2025). Our analysis did not test whether the funding change affected complaint volume, so we treat it only as historical context rather than a causal explanation.

5. Forecasting Complaint Volume

Weekly complaint volume was forecast using SARIMAX and Meta Prophet. The models were trained on historical data from 2022 through 2025 and evaluated using unseen observations from 2026.

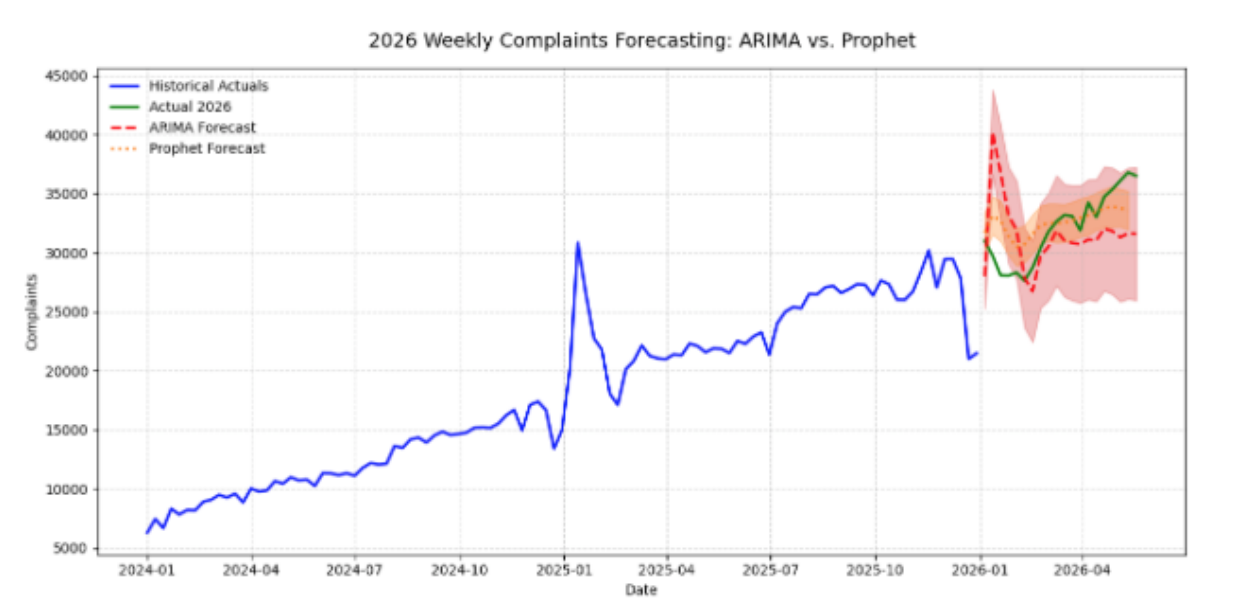
For SARIMAX, auto_arima was used to search model specifications. The auto_arima selected SARIMAX(3,1,1)(0,1,0)[52] based on Akaike's Information Criterion (AIC), where the 52-week seasonal period represented yearly seasonality. Both models forecast the first 20 weeks of 2026.

Evaluation used Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Scaled Error (MASE).

Metric	SARIMAX	Prophet
MAE	3,374.52	1,814.57
RMSE	4,227.78	2,209.33
MAPE	10.67%	5.93%
MASE	0.49	0.26

Prophet had lower error across all four metrics during the 20-week 2026 holdout period. Both models had MASE values below 1, while Prophet performed considerably better relative to the seasonal naive comparison.

Figure 3. Actual 2026 weekly complaints compared with SARIMAX and Prophet forecasts.



These results suggest that forecasting can provide useful information about short-term complaint workload. However, the test period covers only a part of 2026, so the results should not be interpreted as evidence that Prophet will always outperform SARIMAX.

6. Predicting Company Response Outcomes

The final analysis examined whether complaint characteristics could help predict company response categories. Previous research has also applied machine-learning methods to CFPB complaint data to predict company response outcomes and response timeliness (Vaishnav et al., 2024).

This analysis used a different outcome definition from the NLP relief indicator. In the supervised-learning notebook, complaints closed with an explanation or monetary relief were coded as 1, while all other response categories were coded as 0. The code produced 783,169 observations in class 1 and 438,801 in class 0.

We refer to this variable as the company-response target rather than “closed respectfully,” because the latter would imply a subjective judgment that is not captured by the coding rule.

Features included product type, Older American and Servicemember indicators, complaint word count, presence of a dollar sign, severity-related keywords, financial-loss and credit-damage indicators, credit-bureau and credit-union indicators, and whether the complaint came from a Southern state.

This helped connect the different parts of the project: severity features came from the NLP analysis, while the Southern-state feature was motivated by the geographic analysis.

We first fitted a logistic regression model because its coefficients can be converted into odds ratios and interpreted more directly. For example, the estimated odds ratio for Older American was approximately 1.27, meaning that the odds of the class-1 response were 1.27 times those of the reference group while holding the other included predictors constant.

We then compared logistic regression, histogram gradient boosting, and random forest on the holdout data. Because the target classes were imbalanced, Average Precision was used as the main model-comparison metric.

Logistic regression achieved an Average Precision of 0.81, while gradient boosting and random forest each achieved 0.82. The difference was small, so logistic regression remained useful because its predictions were easier to interpret.

We also deployed all three supervised-learning models in a [Streamlit application](#), allowing users to enter complaint characteristics and view the predicted probability of the company-response outcome.

Recent CFPB research also supports the broader idea that characteristics of complaint narratives may be related to outcomes. Shin, Kim, and Shin (2026) analyzed more than 1.1 million CFPB complaints and found evidence that LLM-assisted improvements in complaint presentation were associated with a higher probability of favorable relief. Their research

addresses a different question from ours, but it provides additional evidence that narrative characteristics can contain information related to complaint outcomes.

7. Evaluation, Limitations, and Broader Impact

Taken together, the analyses provide different views of the complaint system. NLP showed that sentiment alone does not capture financial severity. Unsupervised learning identified more specific themes within credit-reporting complaints. Geographic analysis showed why population adjustment matters when comparing regions. Prophet provided the strongest forecasting performance during the 2026 test period, while the supervised models showed that structured and narrative-derived features contain information associated with company response categories.

There are several limitations.

First, the CFPB states that its database is not a statistical sample of all consumer experiences. Consumers who file complaints may differ from consumers who experience financial problems but never report them. Low complaint volume also does not necessarily mean that consumer harm is low. Our findings therefore describe patterns in CFPB complaints rather than all U.S. consumers.

Second, the different analyses used different time periods and subsets. The clustering analysis was also limited to 2025 credit-reporting complaints, and manually assigned topic labels could differ under other models, samples, or researchers.

Third, the severity score used researcher-defined keywords and weights. Sensitivity testing improved confidence in the observed patterns, but the score is not a validated clinical or regulatory measure of consumer harm.

Finally, predictive associations should not be interpreted as causal relationships. A feature associated with a company-response category does not necessarily cause that response.

The main stakeholders affected by this work include consumers, financial institutions, consumer advocates, and regulators. Complaint forecasting and geographic trends could help identify periods or areas that deserve additional attention, while narrative analysis could help surface recurring problems that broad categories may hide.

There is also an important ethical concern. HDBSCAN classified many complaints as outliers. Consumers with unusual or emerging problems should not receive less attention simply because their complaints do not fit a large cluster. A system that automatically prioritizes common topics could unintentionally disadvantage people with rare or complex problems.

For this reason, our models are better suited to supporting human analysis and identifying patterns than making automatic decisions about individual consumers.

8. Incorporating Mentor Feedback

Our mentor suggested that we clarify the project objectives, explain the purpose of the predictive model, define how each part of the project would be evaluated, and be more specific about the role of the visualizations. In response, we developed clearer research questions covering complaint narratives, geographic and temporal patterns, forecasting, and company-response prediction.

We also added evaluation methods for each part of the project. The unsupervised-learning analysis was evaluated using topic interpretability, topic balance, outlier rate, stability across random seeds, Adjusted Rand Index, and sensitivity testing. The forecasting models were evaluated using MAE, RMSE, MAPE, and MASE on unseen 2026 data. For the supervised-learning analysis, we used Average Precision as the main comparison metric because the target classes were imbalanced.

We also clarified the purpose of the predictive analysis. Rather than only trying to predict a company-response category, we used the model to examine whether complaint characteristics and narrative-derived features were associated with company response outcomes. For the visualization component, we used interactive visualizations to explore complaint patterns across geography and time, and we deployed the supervised-learning models in a Streamlit application where users could enter complaint characteristics and view predicted probabilities.

These changes helped us define the goals of each analysis more clearly and connect the methods, evaluation, and final outputs to the main research questions.

9. Conclusion

This project used several data science methods to examine consumer financial complaints from different perspectives. Structured CFPB variables describe important characteristics of complaints, while narrative text provides additional information about consumer experiences that is difficult to represent with a single administrative category.

NLP showed the limitations of general sentiment and provided a domain-specific approach to identifying severity-related language. Unsupervised learning identified 13 detailed credit-reporting themes. Geographic analysis demonstrated the importance of adjusting complaint counts for population. Time-series forecasting showed that weekly complaint volume could be predicted, with Prophet outperforming SARIMAX during the 2026 evaluation period. Finally, supervised learning showed that complaint characteristics and narrative-derived features were associated with company response categories, although more complex models produced only a small improvement over logistic regression.

Overall, the project shows the benefit of using different methods to answer different parts of the same problem. At the same time, the limitations of the data and models show why these methods should support human interpretation rather than replace it.

Statement of Work

Po-Wen Lai contributed to data preparation, exploratory analysis, NLP, VADER sentiment analysis, and financial-severity scoring.

Zhou Jiang contributed to narrative screening, sentence embeddings, UMAP dimensionality reduction, HDBSCAN clustering, stability and sensitivity testing, topic interpretation, and outlier analysis.

Alex Yoon contributed to geocoding, geographic analysis, population normalization, visualization, seasonal analysis, and forecasting using SARIMAX and Prophet.

John Papazian contributed to supervised learning, feature engineering, logistic regression model, gradient boosting model, random forest model, evaluation, interpretation, and Streamlit deployment.

All team members contributed to project planning, interpretation of results, integration of analyses, and preparation of the final deliverables.

References

Consumer Financial Protection Bureau. (n.d.). *Consumer Complaint Database*.

Foohey, P. (2017). Calling on the CFPB for help: Telling stories and consumer protection. *Law and Contemporary Problems*, 80(3), 177–209.

Gao, P., Sun, N., Wang, X., Yang, C., & Zitakis, R. (2023). *NLP-based detection of systematic anomalies among the narratives of consumer complaints*. arXiv. <https://doi.org/10.48550/arXiv.2308.11138>

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 3982–3992.

Schneider, K. E., Tufts, G. C., & Webel, B. (2025). *The Consumer Financial Protection Bureau budget: Background, trends, and policy options* (CRS Report R48295). Congressional Research Service.

Shin, M., Kim, J., & Shin, J. (2026). The adoption and efficacy of large language models in US consumer financial complaints. *Nature Human Behaviour*, 10, 669–680.

Vaishnav, D., Neethinayagam, M., Khaire, A., & Woo, J. (2024). *Predictive analysis of CFPB consumer complaints using machine learning*. arXiv. <https://doi.org/10.48550/arXiv.2407.06399>

Woodard, C. (2025). *The geography of household credit, debt, and delinquency*. Nationhood Lab. <https://www.nationhoodlab.org/the-geography-of-household-credit-debt-and-delinquency/>

The Consumer Financial Protection Bureau Budget: Background, Trends, and Policy Options. Legislation. <https://www.congress.gov/crs-product/R48295>.

Appendix

Appendix A. Variables Used in the Analysis

Variable	Description
Date Received	Date the CFPB received the complaint
Product	Primary financial product or service identified by the consumer
Sub-product	More specific category within the selected product
Issue	Main problem reported by the consumer
Sub-issue	More detailed classification of the reported issue
Consumer Complaint Narrative	Consumer's written description of the complaint when consent for publication was provided
State	State associated with the consumer
ZIP Code	ZIP code associated with the consumer
Company Response to Consumer	Company's recorded response category
Complaint ID	Unique identifier for the complaint

Role of Variables in the Analysis

Analysis Component	Main Variables Used
NLP and Severity Analysis	Consumer Complaint Narrative, Product, Issue, Company Response to Consumer
Unsupervised Topic Analysis	Consumer Complaint Narrative, Product, Issue, Sub-issue
Geographic Analysis	Date Received, State, ZIP Code, Complaint ID
Forecasting	Date Received, Complaint ID

Supervised Classification

Product, Company Response to Consumer, State, demographic tags, and engineered narrative features

Appendix B. NLP Severity Framework

The NLP analysis used VADER to measure the tone of complaint narratives, but sentiment was not treated as a direct measure of complaint severity. Complaints involving fraud, identity theft, or financial hardship could still contain neutral or positive wording. For this reason, sentiment scores were interpreted as indicators of narrative tone rather than consumer harm.

To identify more domain-specific signals, a custom severity framework was developed using spaCy and domain-specific phrase dictionaries.

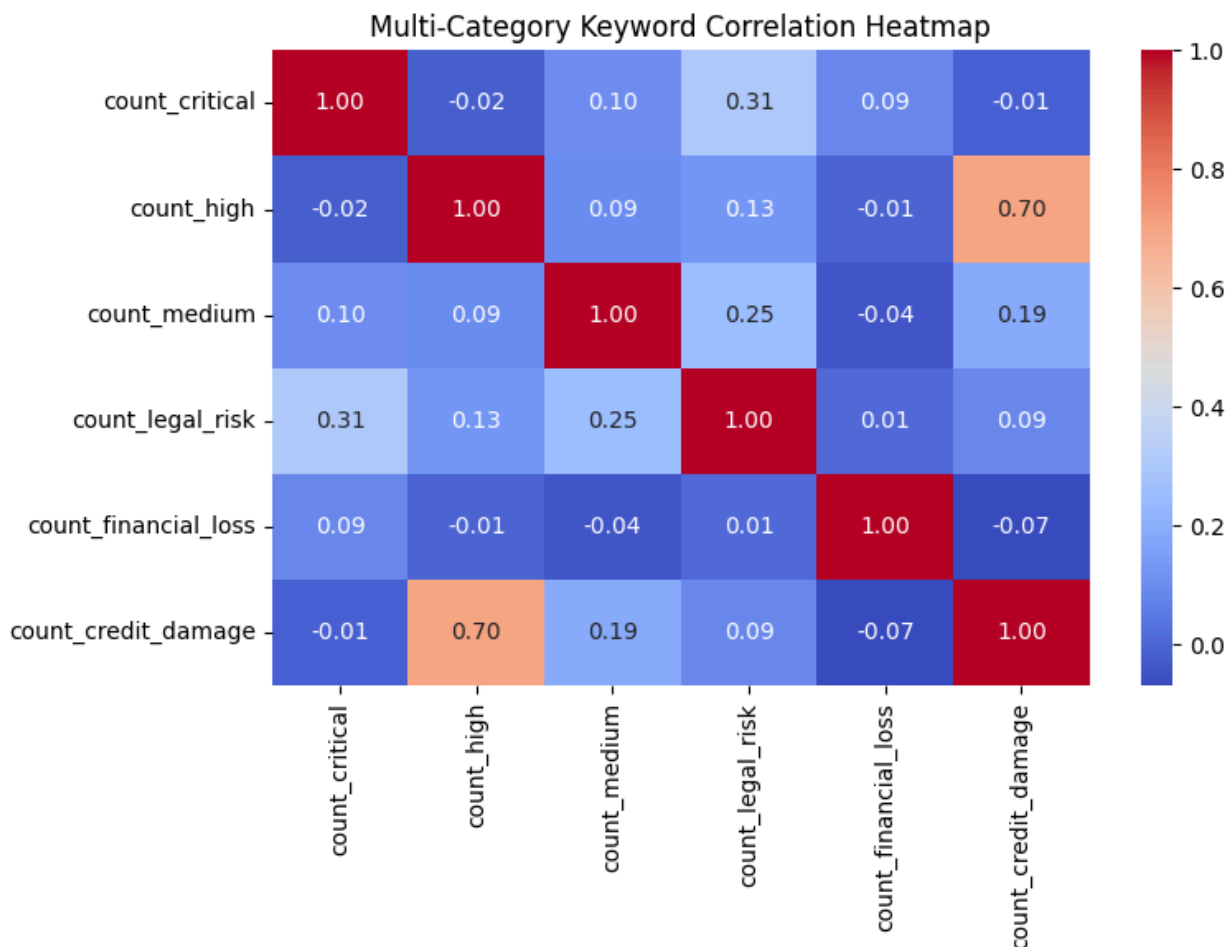
Category	Example Indicators
Critical Severity	foreclosure, bankruptcy, lawsuit, garnishment, eviction, hardship
High Severity	fraud, identity theft, phishing, unauthorized transaction, account takeover
Medium Severity	incorrect reporting, billing error, credit score concern, overdraft fee
Additional Dimensions	legal risk, financial loss, credit damage

The composite severity score was calculated as:

Severity Score = 5(Critical Matches) + 3(High-Severity Matches) + 1(Medium-Severity Matches)

Sensitivity testing with alternative weighting schemes produced similar overall patterns and rankings, suggesting that the main conclusions were not highly dependent on the exact weights used.

The correlation heatmap showed several relationships between complaint severity and narrative language. Critical-severity complaints were more likely to include legal-risk terms, while high-severity complaints were often associated with financial-loss indicators, especially in complaints involving identity theft and fraud. Credit-damage terms were more strongly related to medium-severity complaints than to critical-severity complaints. These results show that different types of financial harm often appear together in complaint narratives, but they should be treated as associations rather than causal relationships.



The NLP analysis also created a relief indicator that was separate from the outcome later used for supervised learning. Monetary or non-monetary relief was coded as 1, while closed with explanation, untimely response, and other outcomes were coded as 0.

CFPB Complaint Word Cloud Insights for Top 10 Complaint Phrases

Analysis of the complaint word cloud highlights the primary drivers of consumer dissatisfaction in 2025, led by core reporting accuracy terms such as *credit*, *report*, *account*, *dispute*, and *investigation*. Fraud and security concerns form another prominent cluster, with terms like *identity theft*, *unauthorized transaction*, *scam*, and *account takeover* emphasizing urgent struggles over stolen funds and reimbursement. Compounding these issues are credit-damage terms—including *delinquent*, *collection*, *derogatory*, and *tradeline*—which point to systemic obstacles in correcting credit bureau records, alongside direct financial loss indicators like *fee*, *balance*, *overdraft*, and *refund*. Operational breakdowns further intensify consumer frustration, as evidenced by recurring service terms like *denied claim*, *failed to respond*, *unresolved*, and *account locked*. Together, these frequency patterns show that credit-reporting disputes, fraud-related language, and servicing problems were prominent themes in the 2025 complaint narratives.

Top 10 Complaint Phrase Words Clouds by Issue



Additional Narrative Findings

Exploratory review identified recurring themes involving inaccurate credit reporting, identity theft, unauthorized transactions, financial loss, dispute handling, and unresolved complaints. Credit-reporting narratives frequently described inaccurate tradelines, falsely reported delinquencies, unsuccessful verification or investigation, and problems related to identity theft.

Appendix C. Unsupervised-Learning Details

The unsupervised analysis focused on credit-reporting narratives. A stratified sample of 10,000 complaints was first drawn from a prepared dataset containing more than one million narratives. After removing exact duplicates, very short submissions, heavily masked records, and narratives with highly repeated openings, 4,727 complaints remained. Of these, 2,902 credit-reporting complaints formed the final modeling dataset.

Each complaint was represented using a 768-dimensional all-mpnet-base-v2 sentence embedding. UMAP reduced these embeddings to five dimensions, followed by HDBSCAN clustering.

The final configuration used:

- UMAP n_neighbors = 10
- Cosine distance
- Random seed = 22
- HDBSCAN minimum cluster size = 50
- HDBSCAN min_samples = 20

Stability and Sensitivity Testing

The selected UMAP configuration produced between 12 and 13 topics across five random seeds. Pairwise Adjusted Rand Index values ranged from 0.716 to 0.773, with a mean of 0.740. Seed 22 was selected because it produced 13 interpretable topics with one of the lower outlier rates.

Replacing cosine distance with Euclidean distance produced only five topics and placed 72.1% of complaints into one dominant cluster. Agreement with the cosine result was low, with an ARI of 0.058.

A second experiment removed common legal phrases and FCRA references before regenerating the embeddings. This model collapsed into two broad topics, with 91.3% of complaints assigned to one dominant cluster. This result suggested that the legal language contained useful information for distinguishing complaint types rather than functioning only as boilerplate.

Topic ID	Interpreted Topic	Complaints
0	FCRA Privacy and Permissible-Purpose Violations	80
1	Unauthorized Hard Credit Inquiries	144
2	General Credit-Report Inaccuracies	112
3	FCRA 605B Identity-Theft Blocking Requests	113
4	Late-Payment Reporting Errors	227
5	Failure to Remove Identity-Theft Accounts	50
6	Identity Theft and Fraudulent Accounts	303

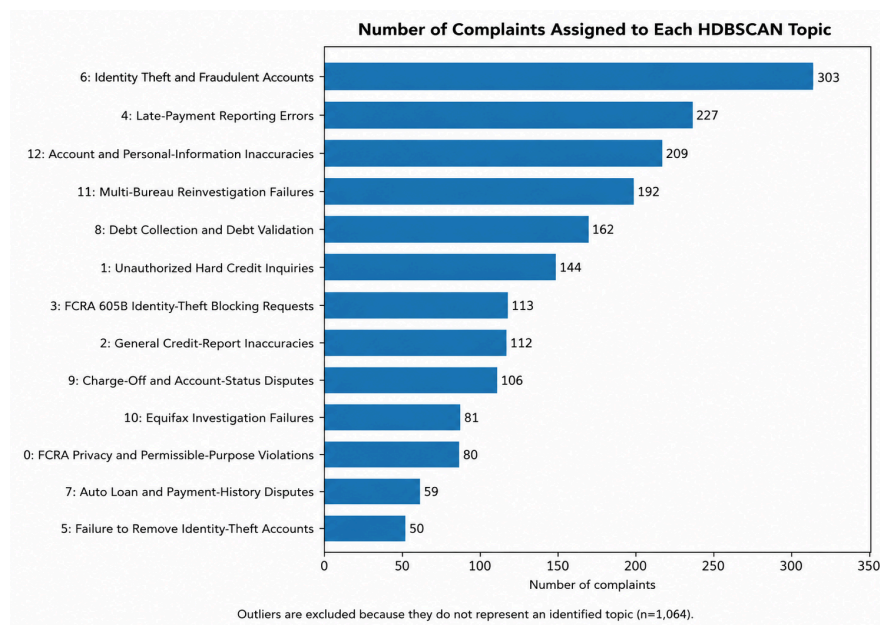
7	Auto Loan and Payment-History Disputes	59
8	Debt Collection and Debt Validation	162
9	Charge-Off and Account-Status Disputes	106
10	Equifax Investigation Failures	81
11	Multi-Bureau Reinvestigation Failures	192
12	Account and Personal-Information Inaccuracies	209

The largest identified topic was Identity Theft and Fraudulent Accounts, followed by Late-Payment Reporting Errors, Account and Personal-Information Inaccuracies, and Multi-Bureau Reinvestigation Failures.

Topic-Count Visualization

The topic-count visualization shows the number of complaints assigned to each of the 13 HDBSCAN topics. Outliers were excluded because they were not assigned to an identified topic. The largest topic contained 303 complaints and the smallest contained 50, showing that the model identified both relatively common and more specific complaint patterns without producing one overwhelmingly dominant topic.

Appendix Figure B1. Complaint Count by HDBSCAN Topic



Outlier Analysis

HDBSCAN classified 1,064 of 2,902 complaints as outliers, corresponding to an outlier rate of 36.7%.

The median narrative length among outliers was 148 words compared with 158.5 words among assigned complaints, suggesting that outlier status was not primarily caused by very short narratives. The median cosine similarity between an outlier and its nearest topic centroid was 0.814, while only 8.7% exceeded a similarity of 0.90.

Rather than assigning these complaints to their nearest topic, the original HDBSCAN outlier labels were retained. Forcing assignments would make the results appear more certain than the density-based clustering supported.

Appendix D. Geographic Analysis Details

The geographic analysis used a 25% sample of CFPB complaints covering 2011 through 2026. Because the data represented a sample, the raw complaint counts in this portion of the analysis should not be interpreted as the complete number of complaints submitted to the CFPB.

ZIP codes were standardized and geocoded using pgeocode. Approximately 3.6 million of 4 million ZIP-code observations, representing 26,418 unique ZIP codes, were converted into latitude and longitude coordinates.

Complaint counts were normalized by population and reported as complaints per 100,000 residents. States were grouped into the Northeast, Midwest, South, and West using U.S. Census regional definitions.

After normalization, the South had a higher complaint rate than the other regions. The analysis did not determine the cause of this difference.

Interactive Geographic Visualization

The project also included an interactive visualization based on a sample of 10,000 complaints. The complete interactive version is available in the project Jupyter notebook.

Appendix E. Forecasting Details

Complaint data were aggregated weekly and analyzed for trend and seasonality before forecasting. Two forecasting methods were compared: SARIMAX and Meta Prophet.

The models were trained on historical observations from 2022 through 2025 and evaluated on observations from 2026 that had not been used for model training.

The SARIMAX specification selected in the project notebook was:

`SARIMAX(3,1,1)(0,1,0)[52]`

The 52-week seasonal period was used to represent annual seasonality in weekly complaint volume.

Prophet produced lower error than SARIMAX on all four metrics during the 2026 evaluation period.

Trend and Seasonal Decomposition

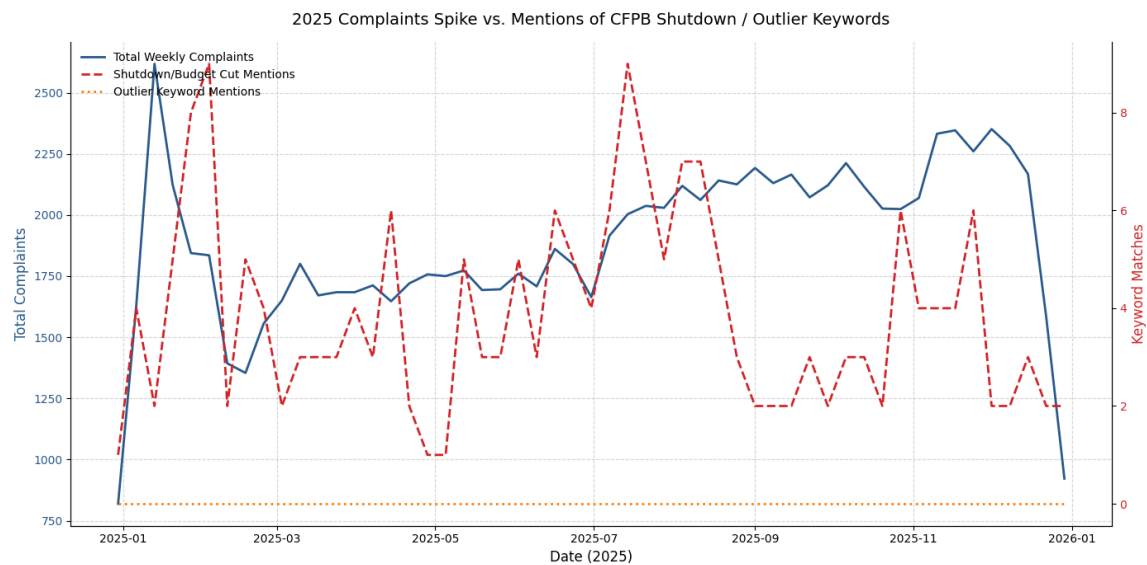
The time-series analysis also examined the longer-term trend and repeating seasonal patterns in complaint volume. Complaint activity generally increased near the beginning of the year and declined toward the end of the year.

Appendix Figure D1. Underlying Trend and Seasonal Patterns in Weekly Complaint Volume



The ARIMA-versus-Prophet forecast comparison is included in the main report because it directly supports one of the project's main findings and does not need to be repeated in the appendix.

Appendix Figure D2. Understanding Cause of Volatility in Early 2025



Using the same filtering and unsupervised learning techniques, we checked for words such as 'defund' and 'cut' in the outliers. The goal was to see if CFPB's defunding announcement was

the main cause of volatility in early 2025, but because ‘defund’ and ‘cut’ were synonymous to financial complaints, we could not find concrete evidence.

Appendix F. Supervised-Learning Details

The supervised-learning analysis combined structured CFPB variables with features engineered from complaint narratives and results from earlier stages of the project.

The final predictor set included:

- Product type
- Older American indicator
- Servicemember indicator
- Complaint word count
- Presence of a dollar sign
- Critical-severity keyword indicator
- High-severity keyword indicator
- Medium-severity keyword indicator
- Financial-loss keyword indicator
- Credit-damage keyword indicator
- Credit-bureau indicator
- Credit-union indicator
- Southern-state indicator

The severity-related features were based on the NLP analysis, while the Southern-state feature was motivated by the geographic analysis. All of the estimated coefficients of the final features in the logistic regression model were statistically significant.

Company-Response Target

The supervised-learning target was different from the relief indicator used in the NLP analysis.

For the supervised model:

Class 1: Closed with explanation or closed with monetary relief

Class 0: Other company response categories

The original supervised-learning work referred to this outcome as "closed respectfully." In the final report, the term company-response target is used because it more accurately reflects the coding rule and does not imply a subjective judgment about the quality of the response.

Logistic Regression Interpretation

The data were divided into a 70% training partition and a 30% holdout validation partition. Logistic regression was used both for statistical analysis and because its coefficients could be exponentiated and thus converted into odds ratios for more direct interpretation.

For example, the Older American indicator had an odds ratio of approximately 1.27, meaning that its estimated odds of the class-1 response were 1.27 times those of the reference group while holding the other included predictors constant.

Complaint word count had an odds ratio of approximately 0.999 for each additional word, which means that each additional word in the consumer complaint narrative slightly decreases the odds of the class-1 response at the margin.

These estimates describe associations within the fitted model and should not be interpreted as causal effects.

Appendix Figure F1. Table of Odds Ratios from the Logistic Regression model

	Odds Ratio	OR Lower CI	OR Upper CI
Product_Checking_or_savings_account	4.860208	4.598196	5.137149
Product_Credit_card	1.819773	1.752289	1.889855
Product_Debt_collection	1.042265	1.020059	1.064954
Product_Debt_or_credit_management	2.483282	2.145865	2.873755
Product_Money_transfer_virtual_currency_or_money_service	16.401459	15.126231	17.784196
Product_Mortgage	7.907559	7.037879	8.884705
Product_Payday_loan_title_loan_personal_loan_or_advance_loan	5.271852	4.635482	5.995585
Product_Prepaid_card	4.710505	3.964658	5.596664
Product_Student_loan	1.150666	1.082529	1.223091
Product_Vehicle_loan_or_lease	3.790955	3.462568	4.150486
Older_American	1.270749	1.205642	1.339372
Servicemember	1.324687	1.288679	1.361702
complaint_word_count	0.999699	0.999672	0.999726
dollar_sign	3.934499	3.854176	4.016497
critical_severity_keywords	1.052210	1.027864	1.077134
high_severity_keywords	1.061304	1.048502	1.074263
medium_severity_keywords	1.094182	1.082980	1.105499
financial_loss_keywords	1.795841	1.629079	1.979675
credit_damage_keywords	1.021774	1.008179	1.035551
credit_bureau	0.299738	0.294015	0.305573
credit_union	0.897164	0.816265	0.986081
Southern_State	0.968500	0.959292	0.977796

Model Comparison

Because the outcome classes were imbalanced, Average Precision was used as the main model-comparison metric.

Model	Average Precision
Logistic Regression	0.81
Gradient Boosting	0.82
Random Forest	0.82

Gradient boosting and random forest achieved slightly higher Average Precision than logistic regression, but the difference was small and the precision and recall performance of the three models was similar. The project also deployed all three Scikit-learn models through a Streamlit application, providing an example of how the predictive work could be translated into an interactive application.

Appendix Figure F2. Streamlit application (with a complaint example) for company-response prediction

<https://mads-capstone-consumer-financial-complaint.streamlit.app/>

MADS Capstone Project: Predicting that a Consumer Financial Complaint is Closed Respectfully

by Alex Yoon, Po-Wen Lai, Zhou Jiang, & John Papazian

Please provide details about your consumer financial complaint submitted to the CFPB:

What type of financial product did you identified in the complaint?

Money transfer, virtual currency, or money service

Are you submitting a complaint by or on behalf of a consumer 62 years or older?

True

Are you submitting a complaint by or on behalf of a servicemember or the spouse or dependent of a servicemember?

True

Are you submitting a complaint about a credit bureau?

False

Are you submitting a complaint about a credit union?

False

Which U.S. state or territory is the complaint being filed?

MI

Please describe what happened and include all details in your complaint

Last week, I tried to send \$100 through CashApp to a friend at the University of Michigan as a graduation gift for the student completing their Master of Applied Data Science degree. However, the transaction did not go through. Nonetheless, CashApp still withdrew \$100 from my account. I want to report this billing error and seek a refund for my financial hardship. Thanks!

Which predictive model to use for scoring?

- ☒ Logistic Regression
- ☐ Gradient Boosting
- ☐ Random Forest

Score using Model

Appendix G. Additional Limitations and Intended Use

The unsupervised topics should not be treated as fixed or objective complaint classifications. Topic labels were assigned manually after reviewing keywords, representative narratives, and

related CFPB fields. Different samples, embedding models, parameter settings, or researchers could produce different topic boundaries.

Complaints classified as outliers should also not be treated as less important or credible. Rare, complex, or emerging problems may not appear frequently enough to form a dense cluster. A system that prioritizes only large clusters could therefore disadvantage consumers with less common problems.

The NLP severity framework is based on researcher-selected terms and weighting values. Although sensitivity testing produced similar rankings, it should be considered a domain-informed indicator rather than a validated measure of consumer harm.

The forecasting models estimate complaint volume, not complaint severity or individual outcomes. Likewise, the supervised models identify predictive associations rather than causal relationships.

For these reasons, the methods developed in this project are most appropriate for exploratory analysis, trend identification, and decision support rather than automated decisions about individual consumers.